bizsim

User guide

Signing in, what each role sees, and what to click in every section: templates, companies, packages, people, sessions and the debrief.

Two jobs, two people: a BUILDER prepares templates, companies and packages; a FACILITATOR starts sessions from a published package and runs the debrief. An ADMIN does both.

A BUILDER cannot run a session; a FACILITATOR cannot edit what they run. The server enforces it, so the sections you can see are the sections your role owns.

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bizsim — user guide

Printable PDF: `docs/bizsim-user-guide.pdf` (regenerate with `python scripts/md_to_pdf.py docs/USER_GUIDE.md docs/bizsim-user-guide.pdf`, needs `fpdf2`).

How to sign in, which section each role sees, and what to click to get a class from "nothing prepared" to "debrief on screen".

The product has **two jobs done by two people**:

Job	Role	Does
Building	BUILDER	prepares templates (the economy), companies, and packages; happens well before a class
Running	FACILITATOR	starts sessions from a published package, seats trainees, opens/computes rounds, runs the debrief
both	ADMIN	everything above
neither	PLAYER	the trainee: signs in, decides, submits

A `BUILDER` cannot run a session. A `FACILITATOR` cannot edit a template, a company, or publish a package. This is enforced by the server, not just hidden on screen — if you cannot see a section, it is the wrong account, not a bug.

1. Signing in

1. Open your site: `https://<your-domain>/` — the browser serves the console and forwards `/api` to the backend.
2. Fill the three boxes on the sign-in card:
   - **Tenant** — your organisation's code, e.g. `demo`
   - **Email** — the address your account was created with (case-sensitive)
   - **Password** — at least 8 characters; the first password is handed to you, change it later
3. Press **Sign in**.

Top left after signing in: a **language selector** (English / Chinese) and **sign out**. The language is saved on your account, so it survives a reload. It translates the interface only — the content you author (product names, event text, trend labels) is never translated.

Forgot the tenant code or have no account? An admin creates accounts (People → Add). If nobody can sign in at all, see §8 "First account" below.

2. What each role sees

Nav item	Builder	Facilitator	Admin
Packages	yes	—	yes
Templates	yes	—	yes
Companies	yes	—	yes
People	yes	yes	yes
Sessions	—	yes	yes

3. Builder — Templates (the economy)

This is where a simulation is designed. There is one shipped template, `standard-v1` ("Standard manufacturing simulation"), marked **global**, and it is **read-only**.

3.1 Make your own copy

1. **Templates** in the left navigation.
2. Find the row **Standard manufacturing simulation** and press **Duplicate**.
3. A copy appears marked **yours** with status **draft** and *Based on* = the shipped key. It opens in the editor automatically; if not, click the row.

Everything below is edited **on your copy**. Press **Save** at the bottom of each tab. A banner confirms *"Saved. New simulations from this template use these values."*

3.2 The six tabs

Settings — the class-level numbers.

Group	Fields
pacing	rounds per simulation, minutes per round, cycles, rounds per cycle
starting position	starting cash, starting brand
costs	fixed cost per round, holding cost per unit, machine cost per worker
brand	brand decay per round
hiring	hiring cost (months), severance (months)
information	market review price, forecast horizon, forecast band
research	min R&D benefit ratio
dynamics	cycle demand drift, cycle cost drift
Per function (table)	salary and starting headcount for Production, Sales, R&D, Admin

Environment — the round-by-round timeline. One row per round: *demand index*, *cost index*, *wage index* (1.00 = normal), *trend strength*, *trend label* (free text, shown to trainees), *note*. Use **Add a row** / **Remove the last row** to change the length, then **Save** (saved whole).
Events — the event deck: surprises drawn once per run.

Column	Meaning
Code	unique short id, e.g. EV-FUEL-SPIKE
Name / Description	what the class sees
From round / To round	the window in which it can be drawn
Chance	probability 0.00–1.00 inside that window
Months	how long the effect lasts (1–12)
Demand ×, Cost ×, Wage ×	multipliers while it is active (1.00 = no change)
Target	one product code, or *all products*

Leave the deck empty for a calm economy. **Save** replaces the whole deck. The server refuses duplicate codes, a probability outside 0–1, `From` > `To`, and non-positive multipliers.
Products — one row per product: code, name, price, unit cost, quality, base demand, units per worker, income elasticity (how much a downturn hits this line), trend affinity.
Channels — code, name, reach weight, brand effect (how much brand helps in that channel).
R&D projects — code, name, target product (or all), advantage type (`UNIT_COST_DOWN`, `CAPACITY_UP`, `QUALITY_UP`, `ATTRACTIVENESS_UP`), size, true cost (hidden from trainees), duration in months, benefit curve (`LINEAR`, `POWER`, `CHANCE`), over-funding handling (`REFUND`, `LOST`).
Note: the product, channel and R&D catalogue **locks once a session has used this template** — those codes are stored in every decision, so rewriting them would strand a live class. Settings, environment and events stay editable (each run freezes its own copy of them).

4. Builder — Companies

A company profile is the reusable definition of a company: its **name** and **which products it carries**. Cash, brand and staff come from the template, not the profile.

1. **Companies** in the navigation.
2. The shipped profile `Standard manufacturer` is **read-only** — press **Duplicate** to get an editable copy, or press **Create profile** to make one from scratch.
3. Edit name, legal name and **product line** (`P1, P2` — or leave empty for the full line), press **Save**.

These profiles are what a package is built from, so make one per company that will compete.

5. Builder — Packages (the hand-over)

A package is the immutable bundle you hand to the facilitator: **one template + the companies**.

1. **Packages** → **Publish a package**.
2. Give it a name used in class, e.g. *Autumn 2026*.
3. Choose the **Template** (your edited copy, not the global one).
4. Tick the **Companies** that will compete.
5. **Publish**.

Publishing **freezes a private copy of the template** and records its version. Editing your draft afterwards changes nothing about this package — to change a published economy, edit the draft and publish a **new** package.

Retire stops new sessions from using a package; sessions already run from it keep their history. **Tidy up** appears when retired packages never ran a session; **Remove N** deletes them (and their frozen copies) after a confirmation. Packages that taught a class are never offered.

6. Facilitator — running a class

6.1 Start a session

1. **Sessions** in the navigation. With no session selected you see **Start a session**.
2. Type a **Session name** (defaults to the package name).
3. Choose the **Package** — the line below shows the competing companies and the template version,

so you can confirm you are running what you think you are.

4. Optional: tick **Trainees**, one per company, in company order. More trainees than companies is refused.
5. **Start session**.

The session opens with one company per profile, its product line already set, and the ticked trainees already seated. If a package is missing, no builder has published one yet.

6.2 The run screen

Header: `Run <n>` · status · `round x of y` · cycles. Buttons:

Button	What it does
Open next round	opens the next round for decisions (round 0 → 1 → …)
Compute round	closes the open round and calculates results for every company
Delete run	removes the session and everything played in it — accounts survive

You can only have **one session in progress at a time**. To run two cohorts in parallel you need two facilitator accounts.

6.3 Seating trainees — Trainees card

Under the run you see one block per company.

- **Seat an existing trainee** → pick an account → **Seat**. This is the normal path: accounts are created once in **People** and re-seated as needed.
- **…or create a new one** — name, email, initial password (8+) → **Create and seat**. This is the walk-in case; the new account outlives the session and can be seated again later.

6.4 Scoreboard

One row per company for the latest computed round: profit, cash, share, stock. Empty until you **Compute round**.

6.5 Debrief

Under the scoreboard, for any computed round:

1. Pick the **Round** in the selector.
2. Read across: revenue, cost, profit, cash, share, lost sales, capacity used/available.
3. **Detail** on a company expands the cost lines: cost of goods sold, payroll, marketing, R&D, fixed cost, stock holding, hiring/firing, capacity, market review.
4. The **Class** footer sums revenue and profit for the whole class.

Every number comes from that round's statements — the debrief cannot disagree with what the engine charged.

6.6 Live

A feed of what has happened in the session (`round.opened`, `decision.submitted`, `round.closed`).

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7. Trainee — playing a round

1. Sign in with the trainee account (tenant + email + password).
2. The decision screen shows the company, **Round x of y**, cash, brand, machines.
3. Fill in:
   - **Production and price** — units to produce and the price per product (stock and unit cost are shown for reference)
   - **Workforce and capacity** — production staff
   - **Marketing** — spend per channel
   - **Research and development** — investment per project, *this month*
   - **Market review** — optional paid report; the cost is deducted from the result
4. **Submit round n**. Submitting again replaces the decision until the round is computed.

Errors come back in your own language and name the field, e.g. *"P1: the price must be greater than zero"* or *"Not enough cash: the plan needs X, you have Y."*

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8. Admin — accounts

People lists every account in the tenant with role and status.

- **Add a trainee** — name, email, initial password (8+), and the **Role** selector: **Trainee** / **Facilitator** / **Builder** / **Admin**. Only an admin sees the role selector; the server refuses staff roles for anyone else.
- **Change a role** — the role cell in the list is a dropdown for admins. Pick the new role and it saves immediately; the account keeps its password and its history. Two guards: you cannot change your **own** role (ask another admin), and the **last active admin** cannot be demoted — create a second admin first.
- **Deactivate** stops a login and keeps their history — use this when a person leaves. **Activate** reverses it.
- **Delete** removes an account that never played. Once someone has a computed decision, deletion is refused with *"deactivate the account instead"* — that is deliberate, history must stay truthful.

First account (fresh deployment)

The server creates the first account from three environment variables on start:

```
BIZSIM_BOOTSTRAP_TENANT       e.g. demo
BIZSIM_BOOTSTRAP_EMAIL        your email
BIZSIM_BOOTSTRAP_ADMIN_SECRET 8+ characters
BIZSIM_BOOTSTRAP_ROLE         optional, defaults to ADMIN
```

Keep it an **ADMIN**: only an admin can create the builder and facilitator accounts. Existing accounts are never rewritten by this; remove the variables once you have signed in.

9. Rules that surprise people

Rule	Why
The shipped template is read-only	**every class would otherwise be playing a different economy**
Duplicate to edit	**your copy is yours; the base stays a reference**
Catalogue locks after a session used the template	**decisions store product codes**
Published packages are frozen	**a class that was prepared cannot be changed mid-term**
Re-publishing makes a new package	**the old one stays as it taught**

Rule	Why
One session per facilitator	a second parallel cohort needs a second account
Sessions are disposable, accounts are permanent	deleting a run keeps the people
A builder cannot see sessions	preparing is not watching

10. If something looks wrong

What you see	Cause / fix
Unauthorized / invalid or expired token	sign out and in again; the token lasts 8 hours
This action is for facilitators	you are signed in as a builder; use a facilitator account for sessions
only a builder prepares…	the account is a facilitator; templates/companies/packages need a builder or admin
No published packages yet	a builder has not published one; see §5
you already have a simulation in progress	finish or delete the current run, or use another facilitator
the shipped profile is read-only	duplicate it first
no Role dropdown when adding someone	you are signed in as a facilitator, or the deployed build predates the selector — an admin sees it; otherwise redeploy (§10, api/version)
created an account with the wrong role	People → the role cell is a dropdown for admins → pick the right role; no need to delete and recreate
this package has been retired	publish a new package

What you see	Cause / fix
Blank screen after a deploy	**check https://<domain>/api/version — latest must match the newest migration; if not, redeploy**

11. Run sheet for a class

Before (builder, days earlier)

1. Duplicate the template, adjust settings / environment / events / catalogue, **Save** each tab.
2. Create or copy the company profiles.
3. Publish a package with those companies.
4. Ask an admin for a facilitator account and trainee accounts (or create them, as admin).

On the day (facilitator)

5. Sign in → **Sessions** → **Start a session** → pick the package → tick trainees → **Start session**.
6. **Open next round** → trainees decide and submit → **Compute round**.
7. Repeat for each round; watch the **Live** feed.
8. **Debrief** → pick the round → walk the class through revenue, cost and profit, expand **Detail**.

After

9. **Delete run** if you want a clean slate (accounts stay), or leave it as history.
10. **Tidy up** on Packages removes retired, never-used packages.